MDF SIGNAL CLEANING AND PEAK TORQUE PREDICTOR

User guide and technical workflow documentation

DOCUMENT SCOPE

This document covers the torque-only signal cleaner, the combined AUTO/CUSTOM torque and temperature cleaner, and the behavior-only Peak Torque Predictor / OverTorque machine-learning pipeline.

Document item	Value
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Application area	Vehicle measurement cleaning and event prediction
Input formats	MDF, MF4, DAT
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1. Project Overview

The project contains two measurement-cleaning applications and one machine-learning pipeline. The cleaners correct faulty torque and temperature signals. The Peak Torque Predictor then identifies driving-behavior windows likely to contain high wheel-torque peaks without using measured wheel torque as an ML input.

Component	Purpose	Main output
Torque-only cleaner	Correct wheel-torque spikes, steps, plateaus, and stuck channels.	Edited MF4, validation plots, and Excel log.
Combined cleaner	AUTO torque cleaning plus CUSTOM temperature-pair cleaning.	Edited MF4, plots, and logs.
Peak Torque Predictor	Predict high-torque peak windows, form events, and validate against real torque.	Model, predictions, event report, scores, and plots.

2. Cleaning Tools Summary

Both cleaners recursively process MDF, MF4, and DAT files from a selected root folder. The output folder is created beside the source folder with the suffix _Edited, and the original source folder is preserved.

2.1 Torque-Only Cleaner

- FWD processes TQ_FL and TQ_FR.
- RWD processes TQ_RL and TQ_RR.
- AWD processes both axles.
- Detected faults include spikes, short faults, steps, plateaus, and long stuck channels.
- Same-signal interpolation is preferred; paired-signal substitution is used when required.

2.2 Combined AUTO/CUSTOM Cleaner

Mode	Behavior
AUTO	Uses the original torque-cleaning workflow and standard wheel-torque channels.
CUSTOM	User selects paired channels for FWD, RWD, or AWD. Current rules are temperature-oriented with a valid range of 15°C to 120°C.

CUSTOM temperature cleaning includes short invalid-value interpolation, local-dip interpolation using a 10-second window and 3-second padding, and full-signal substitution when at least 75% of one signal is invalid while the paired reference has no more than 20% invalid samples.

3. Running and Reviewing the Cleaning Tools

1. Open the required Python file in Spyder or a command prompt.
2. Run the complete .py file.
3. For the combined cleaner, choose AUTO or CUSTOM.
4. Select the root folder containing MDF/MF4/DAT files.
5. In CUSTOM mode, choose the drivetrain and selected channel pairs.
6. Wait for recursive processing to finish.

7. Open <Selected_Root>_Edited and review the MF4, Images_Edited, and Excel log.

Cleaning validation

Compare every 1before and 2after image, confirm that only faulty intervals changed, and verify the output MF4 in the normal measurement-analysis environment before engineering use.

4. Peak Torque Predictor and OverTorque ML Pipeline

The Peak Torque Predictor is a multi-stage machine-learning workflow developed to identify 5-second driving-behavior windows that are likely to contain high wheel-torque peaks. The model does not use the four measured wheel-torque channels as prediction features. Real torque is used to create training labels and later to validate predicted events.

Core modeling principle

Behavior signals are the model input. Wheel torque is excluded from the ML feature matrix. Torque is used only for ground-truth peak detection, label generation, engineering validation, and event scoring.

10.1 Pipeline Components

Stage	Python script	Primary function
1	01_detect_torque_peaks_from_edited.py	Detect physical torque peaks at or above 1800 Nm and create the label source.
2	02A_build_peak_prediction_dataset.py	Convert measurement files into overlapping 5-second windows and calculate non-torque behavior features.
3	03A_train_peak_prediction_xgboost.py	Train and evaluate an XGBoost binary classifier.
4	06_predict_peak_events.py	Apply the saved model to unseen MDF/MF4/DAT files and predict peak windows.
5	09_overtorque_pipeline.py	Run end-to-end inference, merge windows into events, validate against real torque, plot results, and calculate event-level scores.

10.2 End-to-End Concept

```
Cleaned MDF/MF4/DAT measurements
        |
Detect real torque peaks >= 1800 Nm
        |
Create overlapping 5 s behavior windows
        |
Calculate non-torque statistical features
        |
Train XGBoost peak-window classifier
        |
Predict peak windows in unseen files
        |
Merge neighboring positive windows into ML events
        |
Validate ML events against real wheel torque
        |
Create plots, Excel report, CSV, and performance scores
```

5. Stage 1: Real Torque Peak Detection

The first script scans only measurement files located inside folders named Measurements_Edited. It loads available wheel-torque channels, identifies the drivetrain from active torque channels, builds one target absolute-torque signal, and detects local peaks at or above 1800 Nm.

11.1 Input Selection

- Supported formats are MDF, MF4, and DAT, including uppercase extensions.
- Only files whose path contains a folder named Measurements_Edited are accepted.
- Generated before/after files and temporary zoom files are excluded.
- The root folder is configured through ROOT_FOLDER in the script.

11.2 Drivetrain and Target Torque

Detected drivetrain	Active axle logic	Target torque signal
FWD	Front torque active	max(abs(TQ_FL), abs(TQ_FR))
RWD	Rear torque active	max(abs(TQ_RL), abs(TQ_RR))
AWD	Front and rear torque active	max(abs(TQ_FL), abs(TQ_FR), abs(TQ_RL), abs(TQ_RR))
UNKNOWN	No clear active axle	Maximum across all available torque channels

A torque channel is considered active when its maximum absolute value is greater than 50 Nm. If timestamp vectors differ, each selected signal is interpolated onto the first available selected wheel timestamp vector.

11.3 Peak Detection Settings

Setting	Value	Meaning
TORQUE_THRESHOLD_NM	1800 Nm	Minimum detected peak height.
MIN_PEAK_DISTANCE_SEC	0.5 s	Minimum separation between local peaks to avoid repeated samples from one physical peak.
PEAK_PROMINENCE_NM	0 Nm	No additional prominence requirement in the current configuration.

Important distinction

Stage 1 records individual torque peaks. A single peak at or above 1800 Nm is enough to enter the peak table. The final official overtorque-event rule used later is stricter: at least two qualifying peaks within five seconds.

11.4 Stage 1 Outputs

```
_Torque_Peak_Detection_Results/
  All_Torque_Peaks_Above_1800.xlsx
  All_Torque_Peaks_Above_1800.csv
  Peak_Detection_Summary.xlsx
  Peak_Plots/
    <MeasurementFile>_Peaks.png
```

The main Excel workbook contains All_Peaks, File_Summary, Overall_Summary, and Skipped_Files sheets. Each peak row includes the file, vehicle metadata, drivetrain, peak time, peak torque, threshold, used torque channels, file duration, and maximum target torque.

6. Stage 2: Peak-Prediction Dataset Construction

The dataset builder converts every processed measurement into overlapping time windows. Each window is labeled according to whether at least one Stage 1 torque peak occurs inside that window. Torque channels are explicitly excluded from the feature matrix.

12.1 Window Definition

Parameter	Current value	Effect
WINDOW_SEC	5.0 s	Each sample represents five seconds of driving behavior.
STEP_SEC	2.5 s	Successive windows overlap by 2.5 seconds, equivalent to 50% overlap.
MAX_NEGATIVE_WINDOWS_PER_FILE	500	All positive windows are retained, while very long files are limited to 500 negative windows.

12.2 Label Definition

```
peak_label = 1  if at least one detected peak lies inside the 5 s window
peak_label = 0  if no detected peak lies inside the 5 s window
```

The dataset also records the number of peaks in the window and the maximum peak torque, but these label-related columns are metadata and are excluded from model features during training.

12.3 Behavior Feature Channels

Candidate channels cover driver demand, vehicle dynamics, vehicle speed, wheel speed, motor speed, battery behavior, and torque-estimation or torque-request signals. The measured wheel-torque channels TQ_FL, TQ_FR, TQ_RL, and TQ_RR are excluded.

Feature group	Examples
Torque request and control	WhlMotSysTqReq, WhlMotSysFrntTqReq, sVcMtcTc torque-request channels
Driver input	AccrPedlRat, APP_r
Vehicle dynamics	ALgt1, ALat1
Vehicle and wheel speed	Speed, VehSpdIndcdVal, wheel-circumference speed channels
Motor speed	WhlMotSysFrntSpdAct, WhlMotSysSpdAct
Battery and power	HvBattUDc, HvBattPwr, HvBattSoc
Torque estimates	WhlMotSysFrntTqEstTqAct, WhlMotSysTqEst2TqAct

12.4 Statistics per Signal and Window

For every available candidate signal, the builder calculates ten statistics within each five-second window: mean, standard deviation, minimum, maximum, range, absolute maximum, first value, last value, delta, and sample count. Missing signal-window combinations are stored as NaN.

12.5 Dataset Outputs

```
_Peak_ML_Dataset/
  Peak_Prediction_Dataset.xlsx
  Peak_Prediction_Dataset.csv
  Peak_Prediction_Dataset_Summary.xlsx
```

The Excel workbook includes the complete ML dataset, a summary, loaded-channel information per file, and skipped-file details.

7. Stage 3: XGBoost Model Training

The training script loads the CSV dataset, removes metadata and label columns, retains numeric behavior features, removes all-NaN and constant columns, and trains an XGBoost binary classifier.

13.1 Leakage-Resistant Train/Test Split

A GroupShuffleSplit is performed using file_name as the group identifier. With a 25% test size, complete measurement files are assigned to either training or testing. This prevents overlapping windows from the same file appearing in both sets.

13.2 Class Imbalance Handling

The training-set ratio of negative to positive windows is used as scale_pos_weight. Missing values are imputed with the median inside a scikit-learn Pipeline before XGBoost training.

13.3 XGBoost Configuration

Parameter	Value
n_estimators	400
max_depth	4
learning_rate	0.03
subsample	0.85
colsample_bytree	0.85
objective	binary:logistic
tree_method	hist
default prediction threshold	0.50

13.4 Threshold Selection and Evaluation

The script evaluates the default 0.50 probability threshold and also searches the precision-recall curve for the threshold that maximizes F1 on the test set. It reports classification metrics, confusion matrices, ROC AUC, PR AUC, and the best-F1 threshold.

Interpretation caution

The best-F1 threshold is selected on the test set in the current script. This is useful for experimentation, but a separate validation set or nested evaluation should be used before treating the threshold as a final unbiased production setting.

13.5 Training Outputs

```
_Peak_Model_A_XGBoost/
  Peak_Model_A_XGBoost.joblib
  Model_A_Test_Predictions.xlsx
  Model_A_Metrics.xlsx
  Model_A_Feature_Importance.xlsx
```

The model bundle stores the complete preprocessing-and-model pipeline, final feature-column order, target name, default threshold, and best-F1 threshold. Test predictions are classified as TP, FP, FN, or TN and separated into false-positive and false-negative sheets for review.

8. Stage 4: Prediction on Unseen Measurements

The unseen-file prediction script loads the saved model bundle and recreates the same five-second windows and statistical features used during training. Feature alignment is enforced using the feature list stored in the model bundle.

14.1 Prediction Procedure

8. Find MDF, MF4, and DAT files recursively in INPUT_FOLDER.
9. Load available non-torque candidate channels.
10. Build five-second windows with a 2.5-second step.
11. Calculate the same feature statistics used during training.
12. Add any missing trained feature columns as NaN.
13. Calculate peak probability with pipeline.predict_proba().
14. Apply MANUAL_THRESHOLD when provided, otherwise use the stored best-F1 threshold.
15. Save all windows and the subset predicted as positive.

14.2 Prediction Outputs

```
_Peak_Model_A_Predictions/
  All_Window_Predictions.xlsx
  Predicted_OverTorque_Windows.xlsx
  All_Window_Predictions.csv
```

Prediction meaning

A positive prediction means the behavior-only model believes the five-second window contains at least one torque peak at or above 1800 Nm. It does not yet mean that the stricter two-peaks-within-five-seconds event rule has been confirmed.

9. Stage 5: End-to-End OverTorque Pipeline

The final pipeline combines prediction, event formation, torque validation, visual review, event matching, and score calculation in one run. A folder-selection dialog can be used to select the unseen measurement root recursively.

15.1 Current Inference Settings

Setting	Current value	Purpose
WINDOW_SEC	5.0 s	Behavior window duration.
STEP_SEC	2.5 s	Window overlap step.
MANUAL_THRESHOLD	0.40	Overrides the model bundle threshold for window classification.
PLOT_MIN_PROB	0.75	Configured high-confidence plotting threshold; however, the supplied step4_plot implementation currently plots merged predicted events without filtering by this value.
TORQUE_THRESHOLD_NM	1800 Nm	Real-torque validation level.
EVENT_MIN_PEAKS	2	Minimum qualifying peaks for one official real event.
EVENT_WITHIN_SEC	5.0 s	Maximum time span used by the official event rule.
EVENT_MATCH_TOLERANCE_SEC	2.0 s	Tolerance when matching predicted and real event intervals.
BOOTSTRAP_ITERATIONS	2000	File-level bootstrap repetitions for correction-factor intervals.

Implementation note

The code comment says only probabilities at or above PLOT_MIN_PROB are plotted, but the supplied step4_plot function groups and plots every merged predicted event passed to it. If strict high-confidence plotting is required, the event table should be filtered before plotting.

15.2 ML Event Formation

All windows with predicted_label equal to 1 are sorted by file and time. Overlapping or nearby positive windows are merged when the next window starts no later than the current event end plus STEP_SEC. Each merged ML event stores its start, end, duration, window IDs, maximum probability, and mean probability.

15.3 Real OverTorque Event Definition

Official engineering event rule

A real overtorque event contains at least two detected torque peaks, each at or above 1800 Nm, occurring within a five-second span.

The final pipeline scans the complete target absolute-torque signal in each file, detects qualifying peaks, and groups them into non-overlapping real events according to this rule. This is different from the ML training label, where one peak inside a five-second window creates a positive window.

15.4 Torque as Validation Only

Real torque channels are loaded only after prediction. The drivetrain-specific maximum absolute wheel torque is used to check whether predicted events correspond to qualifying physical peaks. Torque never enters the prediction feature matrix.

15.5 Predicted-versus-Real Event Matching

Predicted and real events are matched one-to-one within each file. Interval overlap is allowed with a two-second tolerance around the real event. A matched prediction is TP, an unmatched prediction is FP, and an unmatched real event is FN.

10. Event-Level Metrics and Correction Factor

16.1 Calculated Scores

Metric	Formula or meaning
Sensitivity	TP / (TP + FN). Fraction of real events detected by the ML pipeline.
Precision	TP / (TP + FP). Fraction of predicted events that match real events.
F-score	Harmonic mean of sensitivity and precision.
Correction factor	Real events / predicted events = (TP + FN) / (TP + FP).

16.2 Bootstrap Confidence Intervals

Complete measurement files are resampled with replacement for 2,000 bootstrap iterations. The correction factor is recalculated for each bootstrap sample. The script reports 90%, 95%, and 99% percentile confidence intervals.

Why resample files

Overlapping windows and events from one measurement are not independent observations. Resampling complete files better preserves within-file dependence than resampling individual windows.

11. Final Pipeline Outputs

```
_OverTorque_Pipeline_Results/
  OverTorque_Report.xlsx
  All_Window_Predictions.csv
  Validation_Plots/
    PRED_<MeasurementFile>.png
```

17.1 Excel Report Sheets

Sheet	Content
Model_Scores	Overall TP, FP, FN, sensitivity, precision, F-score, correction factor, and confidence intervals.
Scores_Per_File	Real, predicted, TP, FP, and FN event counts for each file.
Event_Matching	One row for each TP, FP, or FN event match.
Overview	Folders, file counts, windows, events, and thresholds.
Per_File_Summary	Merged and torque-validated event summary by file.
Events_Validated	Predicted events with real-torque validation results.
Real_Events_Per_File	Count of real two-peak overtorque events per file.
Real_Event_Details	Times, peak count, peak values, and maximum torque for real events.
All_Windows	Compact window-level predictions.
Skipped_Files	Files that could not be processed.
Predicted_Events_Per_File	Number of merged ML events per processed file.

17.2 Validation Plots

For each file with predicted events, the pipeline plots the real target torque when available. If torque is unavailable, it attempts to plot a behavior signal such as a torque-request or torque-estimation channel. Predicted event intervals are shaded red and one marker is placed at the highest plotted value inside each event.

12. How to Run the Peak Torque Predictor

18.1 Training Workflow

16. Run the torque peak detector to create the ground-truth peak table.
17. Review Peak_Plots and the Stage 1 summaries.
18. Run the dataset builder to create five-second behavior windows.
19. Check class balance, loaded channels, skipped files, and the feature count.
20. Run the XGBoost training script.
21. Review default-threshold and best-F1 results, false positives, false negatives, and feature importance.
22. Keep the trained joblib bundle together with the scripts that use it.

18.2 Unseen-Vehicle Workflow

23. Set or select the unseen measurement folder.
24. Confirm that the trained model file path is correct.
25. Run 06_predict_peak_events.py for window-level prediction, or run 09_overtorque_pipeline.py for the complete event-level workflow.
26. Review predicted-event plots and the Excel report.
27. Compare ML event counts with real-event counts where torque is available.
28. Review TP, FP, and FN events before interpreting aggregate scores.

18.3 Required Python Packages

```
pip install asammdf numpy pandas scipy matplotlib openpyxl scikit-learn xgboost joblib
```

13. Peak Predictor Validation Checklist

- Confirm Stage 1 reads only the intended Measurements_Edited folders.
- Verify that FWD, RWD, and AWD target torque uses the correct wheel channels.
- Inspect peak plots to confirm the 1800 Nm detections are physically correct.
- Confirm measured wheel-torque channels are absent from ML feature columns.
- Confirm training and test sets contain different file groups.
- Review positive and negative window counts for class imbalance.
- Verify unseen files provide enough of the trained feature channels.
- Review the probability threshold used in each run.
- Inspect predicted windows before merging and merged events afterward.
- Check real-event calculation uses at least two peaks within five seconds.
- Review event-matching tolerance and one-to-one matches.
- Inspect false positives and false negatives individually.
- Treat aggregate metrics together with per-vehicle and per-driving-profile results.

14. Known Limitations and Maintenance Notes

Area	Current limitation or maintenance requirement
Paths	ROOT_FOLDER, INPUT_FOLDER, dataset path, and model path are hard-coded in several scripts unless a folder dialog is used.
Feature availability	New vehicles may not contain all trained channels. Missing feature columns become NaN and are median-imputed by the trained pipeline.
Channel names	Suffix and contains matching are flexible but can choose an unintended channel when several similar names exist.
Training label versus event rule	The classifier learns one-or-more peaks per five-second window, while the final engineering event requires at least two peaks within five seconds.
Threshold tuning	The best-F1 threshold is selected on the test set in the training script; use separate validation for unbiased final tuning.
High-confidence plotting	PLOT_MIN_PROB is configured but not currently applied inside step4_plot.

Area	Current limitation or maintenance requirement
Drivetrain metadata	Some scripts detect drivetrain from torque activity, while others detect it from file paths. Folder naming and channel activity should be consistent.
Event matching	Results depend on the two-second tolerance and the one-to-one greedy matching order.
Generalization	Performance should be reported by unseen vehicle, drivetrain, and driving profile, not only as one combined number.

Recommended release practice

Version the peak table, dataset, feature list, trained model, probability threshold, event rule, and evaluation folder together. A model result is reproducible only when all of these artifacts are linked to the same release.

15. Peak Predictor Quick Reference

Item	Current definition
ML prediction target	Does this five-second window contain at least one torque peak >= 1800 Nm?
ML inputs	Non-wheel-torque behavior statistics only.
Window duration and step	5.0 seconds with a 2.5-second step.
Model	XGBoost binary classifier inside a median-imputation pipeline.
Default training threshold	0.50, with an additional best-F1 threshold reported.
Final-pipeline manual threshold	0.40 in the supplied configuration.
Official real event	At least two torque peaks >= 1800 Nm within five seconds.
Torque role	Ground-truth labels and validation only, never ML input.
Primary final report	_OverTorque_Pipeline_Results/OverTorque_Report.xlsx

Updated Project Summary

The documented project now contains three connected engineering capabilities: a torque-only cleaner, a combined AUTO/CUSTOM torque and temperature cleaner, and a peak-torque prediction pipeline. The cleaning tools improve measurement quality. The peak predictor then uses behavior-only five-second windows to estimate where high-torque peaks are likely, merges positive windows into events, and validates those events against real wheel torque using the official two-peaks-within-five-seconds definition.